

Project Demonstration

1. Overview

This document specifies the demonstration flow, scenarios, and outputs of the HDI Prediction System, illustrating its prediction accuracy and user interface features.

2. Core Demonstration Scenarios

To showcase the system's capabilities during evaluations, we present three distinct scenarios that demonstrate the prediction and classification pipeline.

Scenario 1: Predicting Very High Development (Developed Nations)

- **Objective:** Simulate developmental indicators matching a highly developed economy (e.g., strong educational systems and high national income).
- **Parameters:**
 - Country: `Country\_A`
 - Life Expectancy: `82.0 years`
 - Expected Years of Schooling (EYS): `16.0 years`
 - Mean Years of Schooling (MYS): `13.0 years`
 - GNI per Capita: `\$55,000`
- **Model Output:** Predicted HDI = `1.0` (Clamped maximum).
- **Visuals:** Displays a green badge (🟢 Very High) and maps the score to the upper limit of the visual gauge.
- **Insights:** Recommends advanced research investments and technology sector support.

Scenario 2: Identifying Development Gaps in Emerging Economies

- **Objective:** Model a developing nation profile with moderate health and education metrics.
- **Parameters:**
 - Country: `Country\_D`
 - Life Expectancy: `68.0 years`
 - Expected Years of Schooling (EYS): `11.0 years`
 - Mean Years of Schooling (MYS): `7.5 years`
 - GNI per Capita: `\$12,000`
- **Model Output:** Predicted HDI = `0.7245`.
- **Visuals:** Displays a blue badge (🟡 High Development) and centers the score on the progress gauge.
- **Insights:** Recommends expanding secondary school enrollment and supporting local industries.

Scenario 3: Assessing Low Development (Intervention Planning)

- **Objective:** Evaluate a low-development profile that requires direct policy interventions.
- **Parameters:**
 - Country: `Country\_C`
 - Life Expectancy: `52.0 years`
 - Expected Years of Schooling (EYS): `5.0 years`
 - Mean Years of Schooling (MYS): `3.0 years`
 - GNI per Capita: `\$1,500`
- **Model Output:** Predicted HDI = `0.5408` (Below 0.550 threshold).
- **Visuals:** Displays a red badge (● Low Development) and fills only the bottom portion of the progress gauge.
- **Insights:** Recommends basic vaccination coverage, primary education funding, and humanitarian aid coordination.

3. UI Features & Design Choices

- **Dark Gradient Backdrop:** Uses a dark-mode theme to enhance visual appeal.
- **Frosted Glass Cards:** Features a modern glassmorphic look to organize input forms and results.
- **Dynamic CSS Gauges:** Progress meters map predicted scores to fill widths dynamically.
- **Color-Coded Badges:** Color coding (green, blue, orange, red) matches standard UNDP categories.